4.1.2 Balancing exploration and exploitation

In this subsection, we discuss one of the problems inherent in any decision-making problems, including BayesOpt, which is the balance between sufficient exploration of the entire search space and timely exploitation of regions that yield good results. This discussion will help us form an idea of what a good BayesOpt policy should be doing, and know to look out for how each of the policies we will be learning about addresses this tradeoff.

As an example of the exploration–exploitation tradeoff, imagine you are dining at a restaurant that you have only been a few times before. You know that this restaurant has great burgers but you're not sure if their fish and steaks are any good. Here, you are faced with a exploration-exploitation problem, where you need to choose between ordering something new that might be an excellent dish (**exploration**) and sticking with what works (**exploitation**).

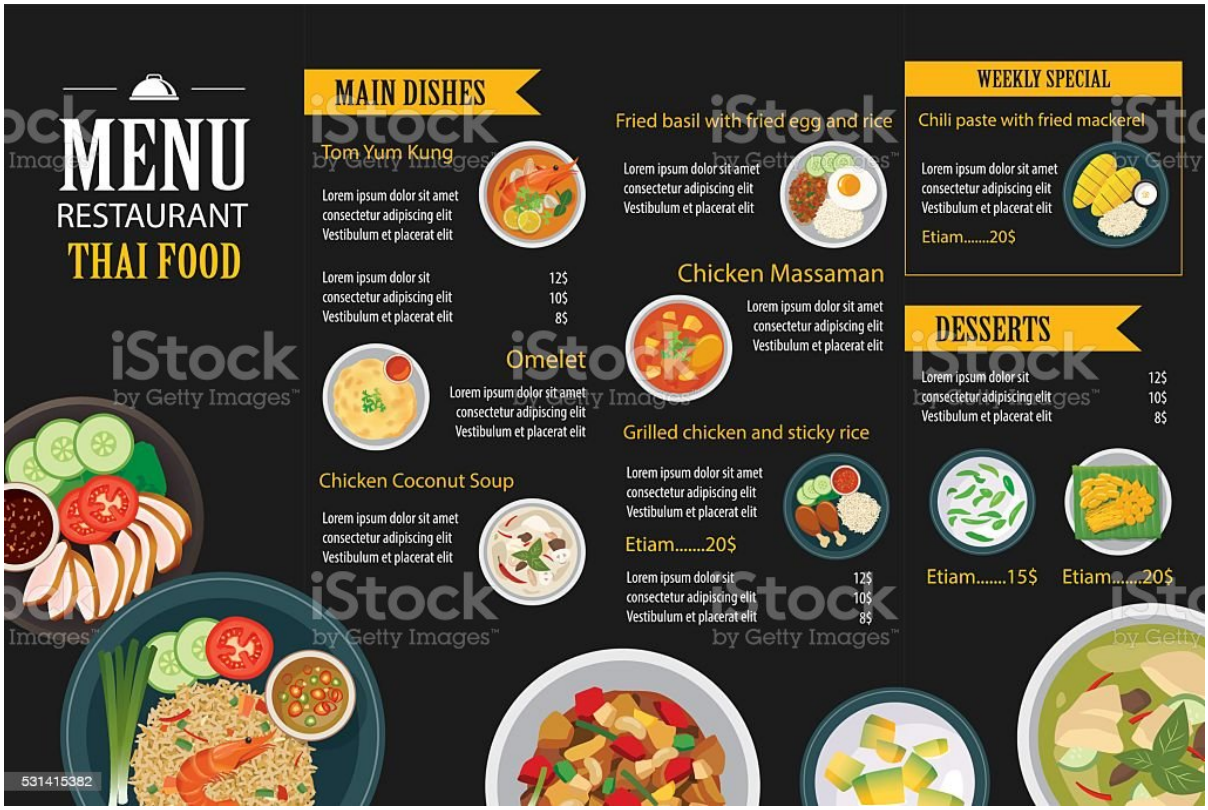


Figure 4.5 Ordering from a menu at a restaurant has an inherent exploration (trying something new) vs. exploitation (sticking to what works) tradeoff.

Excessive exploration might force you to order something you don't like, while constant exploitation runs the risk of missing out on a dish that you will really like. Therefore, a reasonable balance between the two is crucial.

This ubiquitous problem is found not only in ordering food but also in common problems in artificial intelligence such as reinforcement learning, product recommendation, and scientific discovery. In BayesOpt, we face the same tradeoff: we need to sufficiently explore the search

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